

Controllable Object Addition for Text-guided Image Editing using Spatial Conditions

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Outline

- ① Task
- ② Related Work
- ③ Method
- ④ Expected Results

01 Task

Motivation

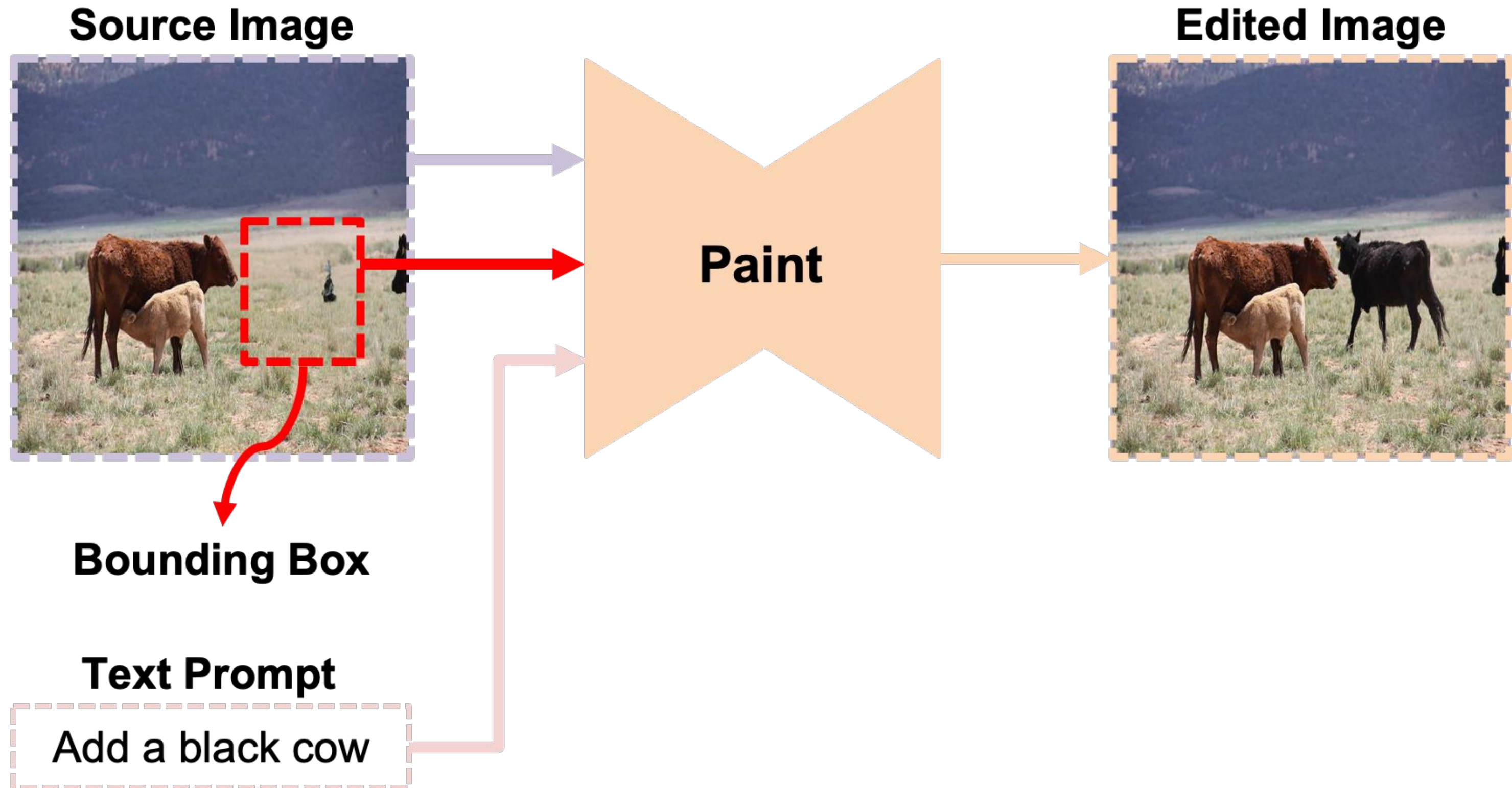
Existing Progress

- Diffusion-based image editing can realistically add objects from text prompts.

Current Limitation

- Object placement is automatically determined by the model.
- Users cannot explicitly control location or scale.

Goal: bounding-box guidance for controllable object addition



02 Related Work

Related Work: Image Editing Methods

Mask-Based Editing

- Uses a mask to specify the edit region
- Limitation: requires user-provided masks

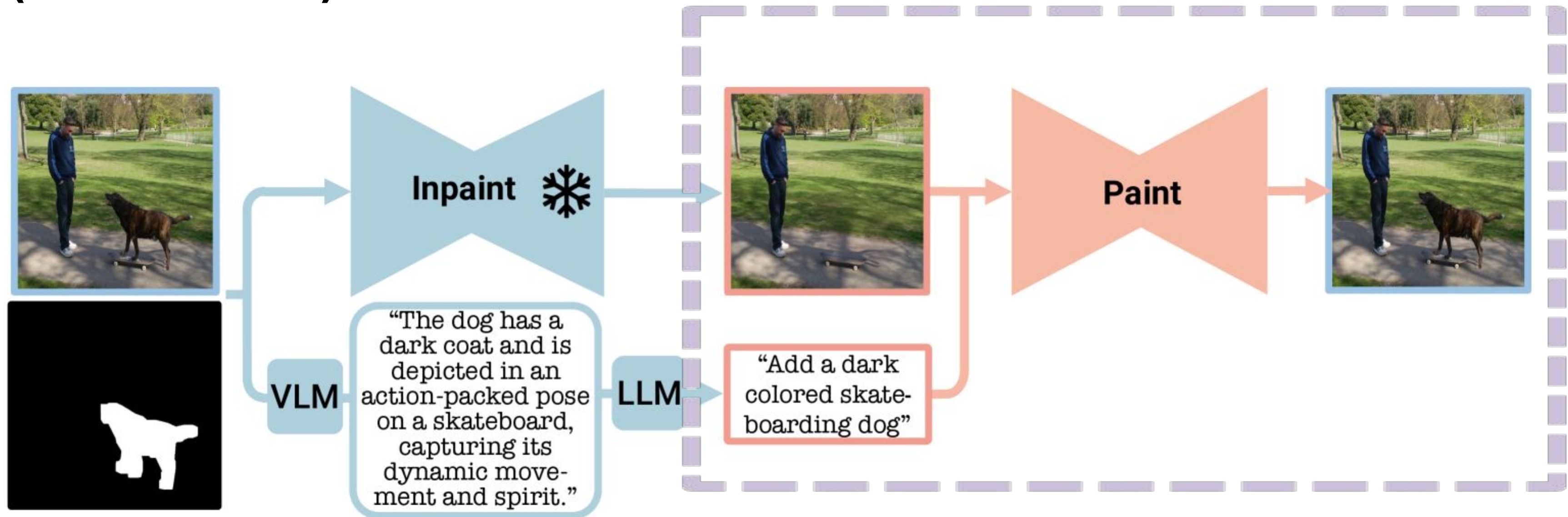
Mask-Free Editing

- Uses text to edit images without masks
- Limitation: local edits may change unwanted regions

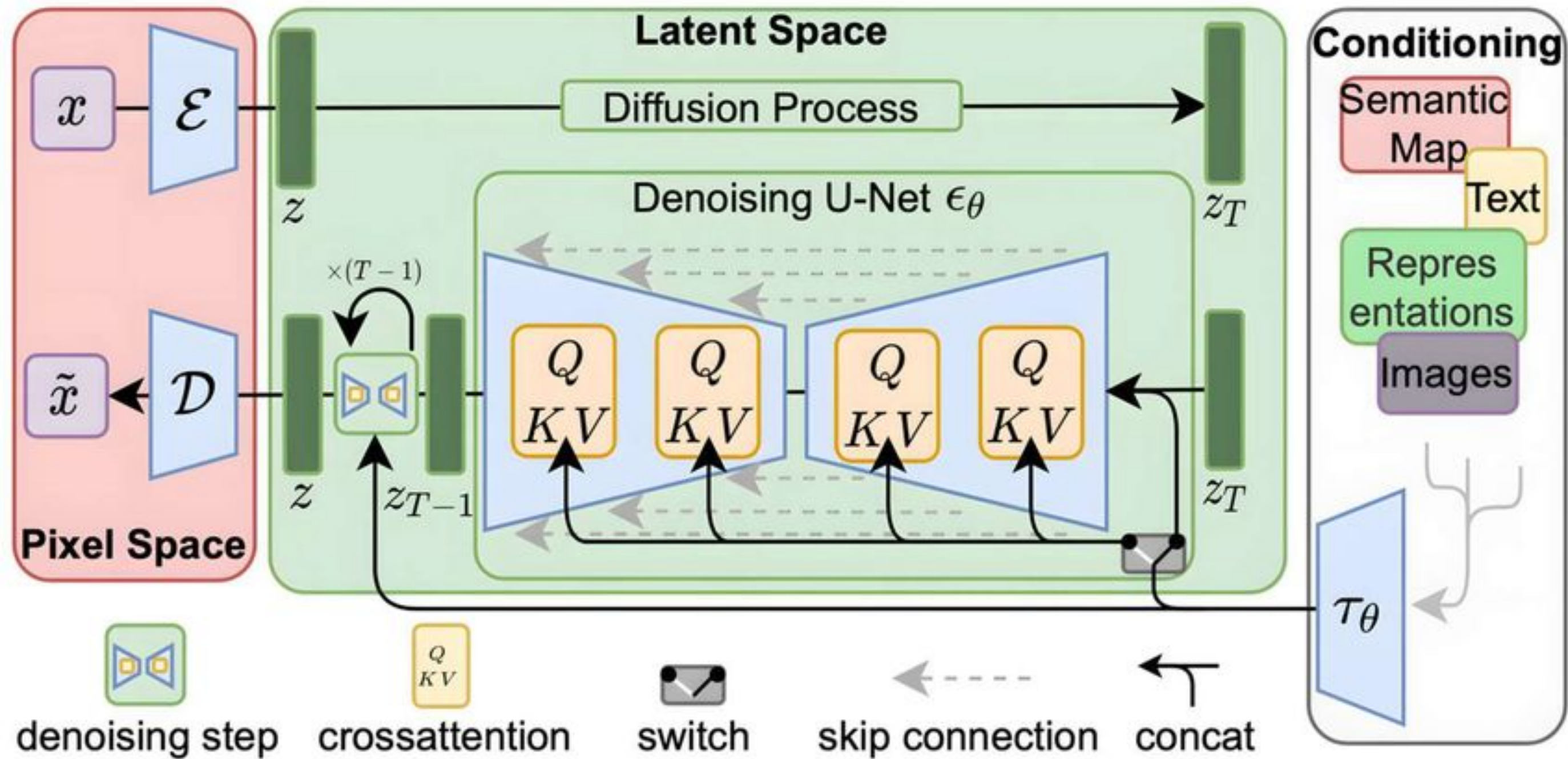
Instruction-Based Editing

- Uses natural language instructions for example “Add a dog”
- Limitation: instructions may lack fine-grained object details

Paint by Inpaint: Learning to Add Image Objects by Removing Them First (CVPR 2025)

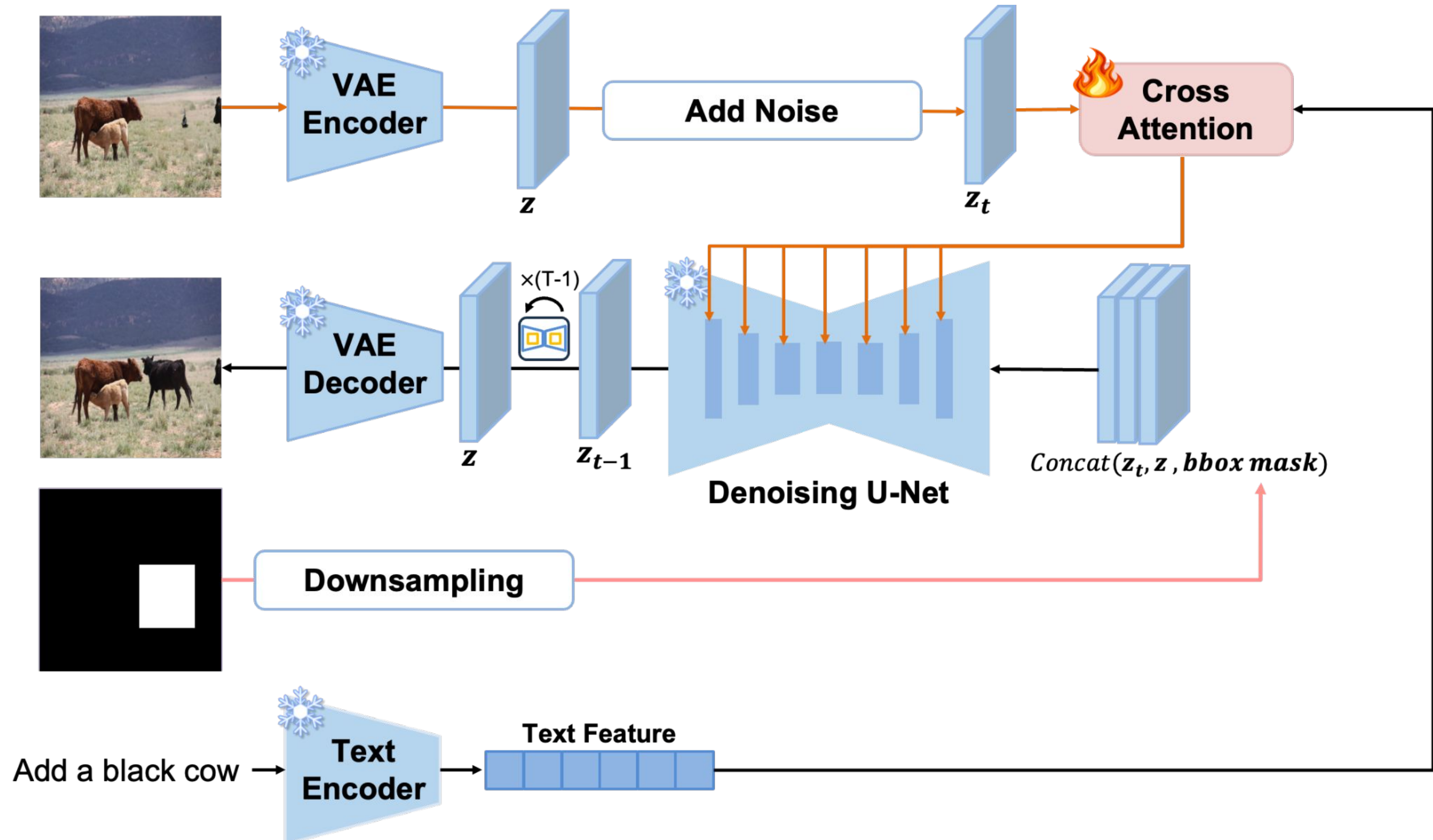


Base model: Stable Diffusion 1.5

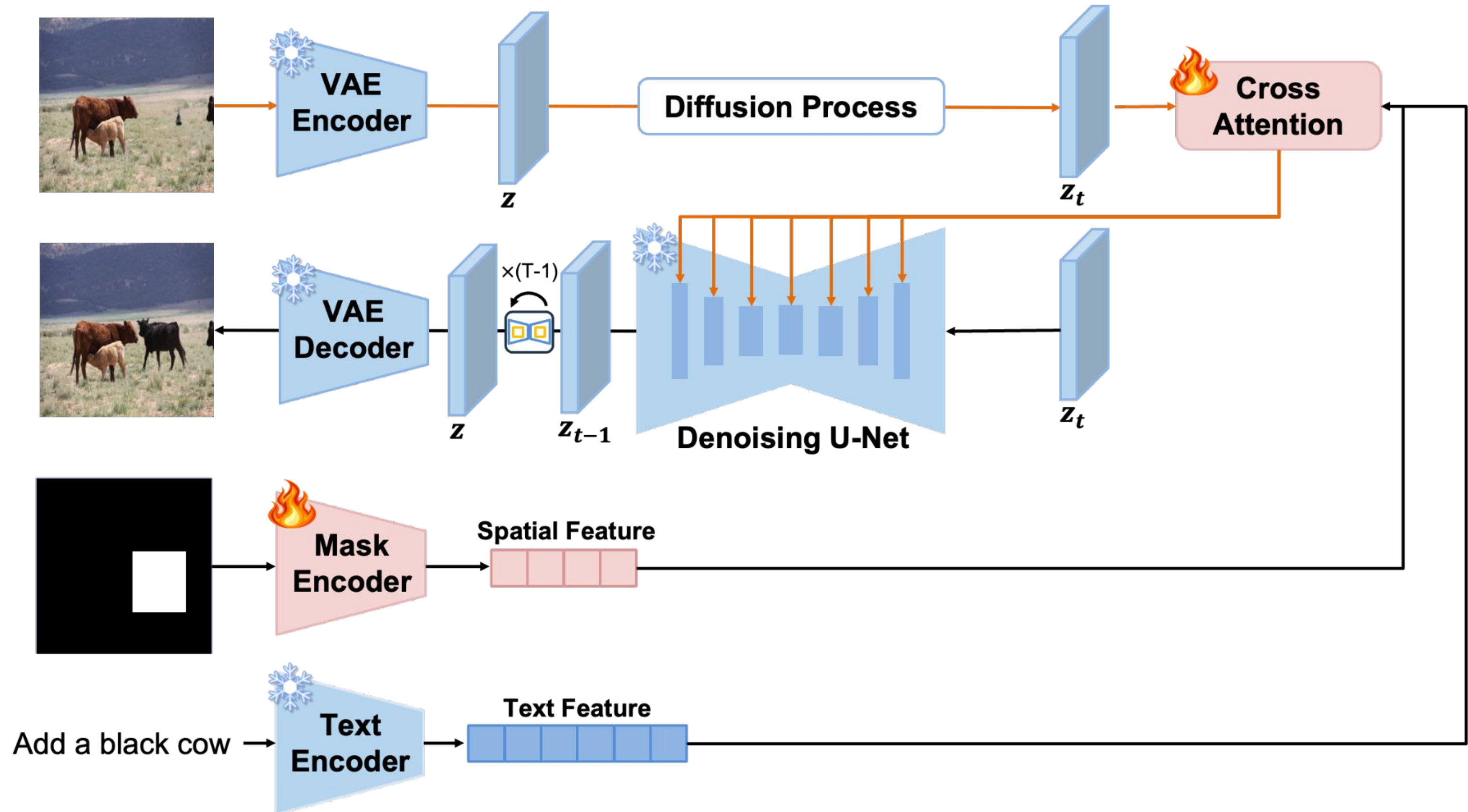


03 Method

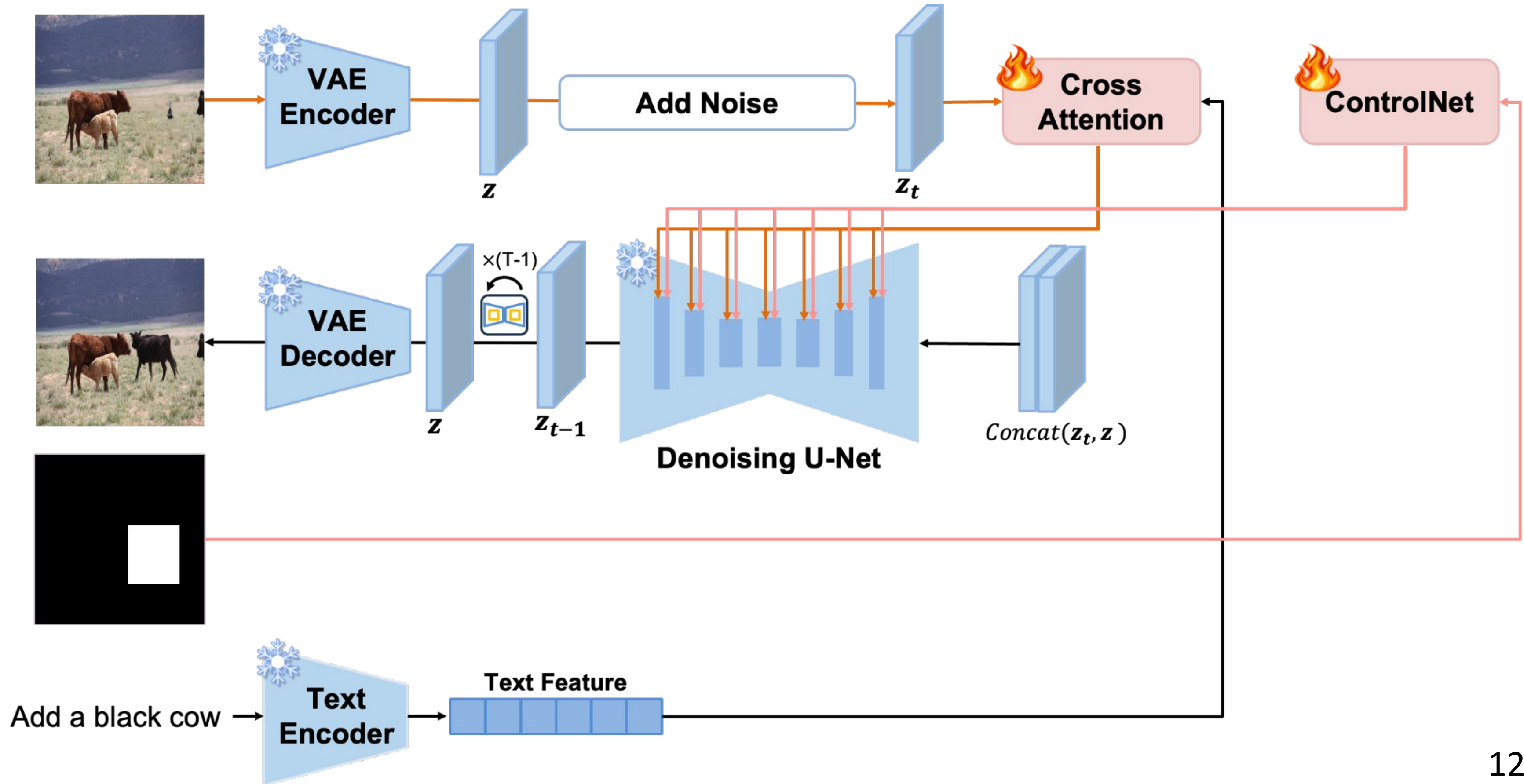
Strategy A: Input-level Conditioning



Strategy B: Attention-level Conditioning



Strategy C: Feature-level Conditioning



Dataset

- **Training / Validation Dataset**

PIPE Dataset

Used to train the proposed BBox-guided editing model.

- **Benchmark Dataset**

MagicBrush Dataset

Used to evaluate generalization ability on real-world editing tasks.

04 Expected Results

Benchmark

Category	Metric	Purpose
Semantic Alignment	CLIP Score ↑	Follows text instruction
Spatial Control	Edit-region IoU ↑	Edits occur in intended region
Background Preservation	Outside LPIPS ↓	Preserve non-edited area
Object Quality	Inside LPIPS ↓	Generated object matches target

Q & A
